Specialised Cells Worksheet

Many organisms are multi-cellular - they are made up of lots of cells, not just one! Many of these cells are specialised, sharing out the *life processes* (they work together as a team, supporting the organism). Specialist cells occur in both animals and plants…

Animal Cells

Draw out the shapes of some of the specialised animal cells you have seen

Animal Cell	Cell Diagram	Function	Specialisation
Red blood cells		To carry oxygen	• Large surface area for oxygen • Haemoglobin which binds the oxygen
Nerve cell		To carry nerve impulses to different parts of the body	• Long • Connections at each end • Can carry electrical signals
Female reproductive cell		To join with a male cell, and then to provide food for the new cell that has formed	• Large • Contains a large amount of cytoplasm
Male reproductive cell		To reach the female cell in order to join with it	• Long tail for swimming • Head which is able to penetrate the female cell

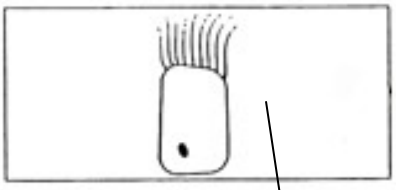
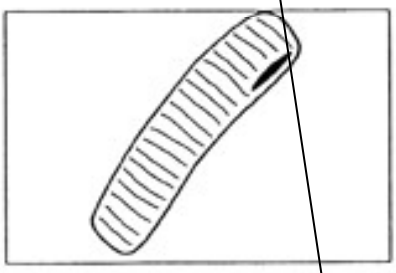
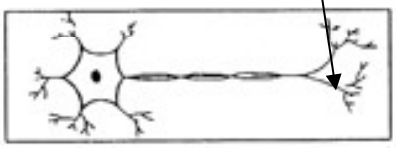
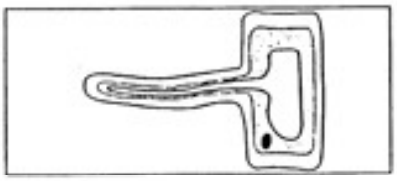
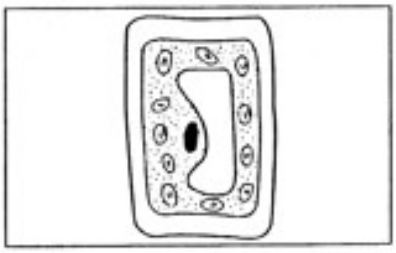
Plant Cells

Draw out the shapes of some of the specialised plant cells you have seen

Animal Cell	Cell Diagram	Function	Specialisation
Root hair cells		To absorb water and minerals	• Large surface area
Leaf cell		To absorb sunlight (needed for photosynthesis)	• Large surface area • Many chloroplasts (containing chlorophyll, for photosynthesis)

Cells & Functions

Match up the picture with the name of the cell, followed by its function…

cell	name	job
	muscle cell	strands wave about to move things
	root hair cell	it can change length
	cell with cilia	it takes water from the soil
	palisade leaf cell	it carries messages
	nerve cell	it has lots of chloroplasts to help the plant make food